

Lilygo Pager: Remote Switch (Relay)

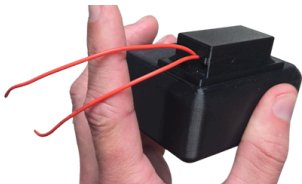
The Blackout Comms off-grid messaging & location firmware is capable of working with several add-on hardware components. The single relay from Sparkfun is a compatible module.



The pager can remotely control a circuit and is capable of messaging/location/meshing.

If you don't understand what this is our how it works, DO NOT BUILD IT!

Be aware this is for experimental & educational purposes only! This is a fun/novelty feature of Blackout Comms, but can be dangerous if you don't know what you're doing. Blackout Comms may contain bugs that cause the circuit to close/open more than once, so be aware of that as well.



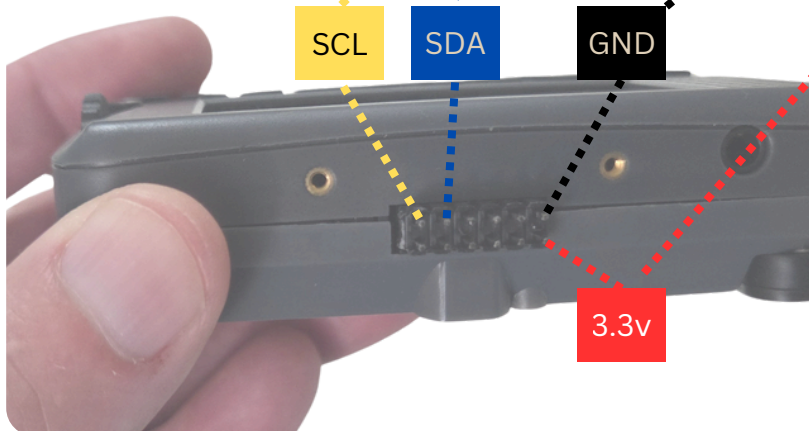
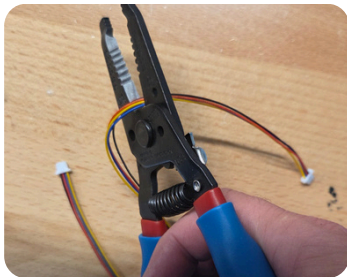
FIRST: Install Blackout Comms

Download Blackout Comms for Lilygo Pager are at:
<https://www.chatters.io/pager>



Pager Expansion Port & Header Pins

You will need to create an adapter between the pager expansion port and a standard QWIIIC cable. Once this is done, the pager is ready to take Blackout Comm compatible modules.



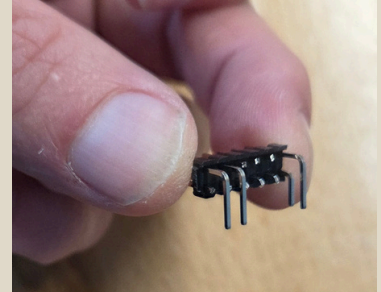
Prepare a set of header pins that match the Pager



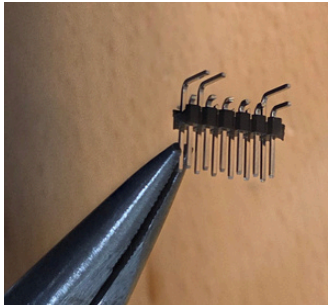
Using a 90° double-row pin header, trim it to 6x2.

Trim all the unused pins so they're not in the way.

We only need SDA, SCL, GND and 3.3v.



Solder the QWIIC Wires to the Header Pins



Use heat shrink insulator and/or liquid electrical tape to insulate the pins and bare wire, so they can't come in contact with one another.

Install the 4 Knurled Nuts

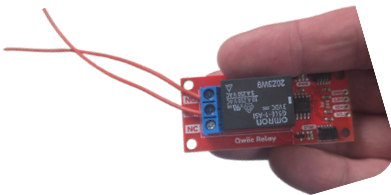


The printed backpack/housing has 6 holes for heat sinking M2 knurled nuts into.

The easiest way to sink these is with a soldering tip for melting knurled nuts into place.



Connect External Relay Wires



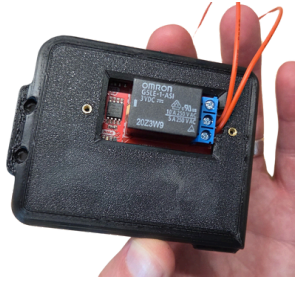
Attach two separate wires to the *Normally Open & Common* terminals of the relay.

These wires are what you'll insert into a circuit you want to be able to open/close

Connect the Header & Relay

- Install the knurled nuts into the printed enclosure
- Connect the relay to your header pin adapter
- Install the relay into the enclosure





Mount the the Relay Into Place

Ensure the relay is not pressing into the back of the pager.

Secure the relay with M2 screws (only 2-3 screws are necessary).



Plug In Header Pins

Plug the header pins into the pager, so the pins are angled down (away from the screen) as shown here.



Mount the Backpack Onto the Pager



Place the backpack over the pager, careful not to pinch or trap any wires.

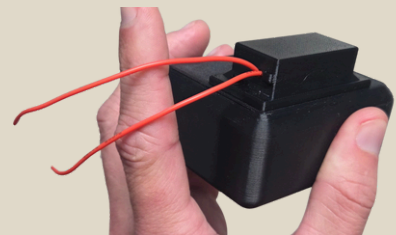
The relay & relay wires should be outside the back of the case as shown.

Use 6 M2 screws to secure.

Install the Relay Cover

Push the relay wires through the relay cover, so you can access them any time. You can zip tie them if you're not using them.

No current flows through these wires unless you supply the current via a separate circuit.



Parts List

To assemble using the instructions above, you'll need soldering equipment & supplies, as well as the components listed below:



Lilygo Pager

[Rokland](#), [Lilygo](#)

Only compatible with SX1262 Models



SparkFun QWIIC Single Relay

[Sparkfun](#)



QWIIC Cable

[Amazon](#)



M2 Knurled Nuts

[Amazon](#)



M2 Screws

[Amazon](#)



90 Degree 2 Row Header Pins

[Amazon](#)